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Ripples in the Reserve Pool: Unveiling Liquidity Signals from the Federal Funds Market

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I. Introduction

When the Federal Reserve shifted to an ample-reserves framework, many observers started to question the relevance of the fed funds market. With reserves ample or abundant and rates anchored by administered floors, interbank borrowing declined sharply, and the market appeared to serve mainly as an arbitrage venue for non-bank institutions. But new research by Anbil, Infante, and Senyuz (2026) demonstrates that the fed funds market retains substantial diagnostic value—if you know where to look. The key insight is that different types of fed funds transactions reveal different information: arbitrage trades by non-bank lenders tell us little about bank liquidity, while lending between banks provides real-time signals about distributional frictions that aggregate reserve measures cannot capture. This note explains why these distinctions matter for monetary policy implementation.

When Federal Home Loan Banks (FHLBs) lend to foreign banks, spreads remain compressed and elastic—revealing little about bank liquidity. But when domestic banks lend reserves to one another, rates become highly sensitive to lenders' reserve positions and the distribution of liquidity across the banking system. This bank-to-bank segment acts as an early warning system for distributional stress, even when aggregate reserves appear ample. These insights are likely to become more important if there are less reserves in the banking system. In a lower reserves environment interbank activity should increase, strengthening the role of the fed funds market as a barometer for liquidity conditions and frictions related to the distribution of liquidity across the banking system. In a potentially more dynamic fed funds market, the price discovery process provides valuable signals about the distribution of liquidity. Monitoring these dynamics is therefore important for assessing financial conditions and the transmission of monetary policy.

II. The Evolution of the Fed Funds Market from Pre-2008 to the Shift to an Ample-Reserves Framework

The fed funds market is an interbank market where depository institutions lend reserve balances to other depository institutions overnight on an uncollateralized basis. Prior to the 2008 global financial crisis (GFC), the fed funds market served as the primary venue for interbank lending and borrowing. It played a key role in implementing monetary policy through reserve

management in an environment of scarce reserves and reserve requirements, and acted as a vital indicator of overall liquidity in the banking system. The Fed used open market operations to inject or drain reserves to control the federal funds rate—the Fed's main monetary policy tool—, which in turn affected broader economic conditions. Thus, the fed funds market was at the heart of U.S. monetary policy implementation, serving as both a tool to execute the Fed's monetary policy stance and a barometer of financial system health. Its functioning provided key insights into liquidity conditions and played a central role in the broader interest rate environment.

The 2008 financial crisis prompted a significant expansion of the Fed's balance sheet through large-scale asset purchases that led to a substantial increase in reserve balances. Just prior to that expansion, the Fed introduced interest on excess reserves which allowed the Fed to remunerate banks' reserve balances above their required level. With the expansion of the balance sheet, in which the fed funds trading was no longer affected by the supply of reserves, interest on reserve balances (IORB) became the new policy tool to maintain rate control. These changes in the implementation framework fundamentally altered the dynamics of the fed funds market. Overall trading volumes declined significantly in the following years accompanied by a shift in participant composition within the fed funds market.

Today, the fed funds market is characterized by the dominant role of non-bank lenders, primarily Federal Home Loan Banks (FHLBs) who are natural lenders in this market because their reserve balances are not remunerated. Over the years following the GFC, U.S. branches of foreign banks have emerged as the primary borrowers of fed funds, to engage in arbitrage: Since these foreign banks face fewer regulatory constraints, they routinely borrow at rates below IORB from FHLBs to earn the spread between fed funds and IORB.[1]

In the current environment, domestic banks participate in the fed funds market intermittently mainly for liquidity management purposes. Notably, "bankers' banks" have become key intermediaries for smaller institutions, facilitating their access to wholesale funding markets.

III. Heterogeneity in the Fed Funds Market

The current structure of the fed funds market exhibits significant heterogeneity among both lenders and borrowers. Considering this heterogeneity and its implications are crucial for understanding the dynamics of the fed funds market and interpreting the information it provides about liquidity conditions in the banking system.

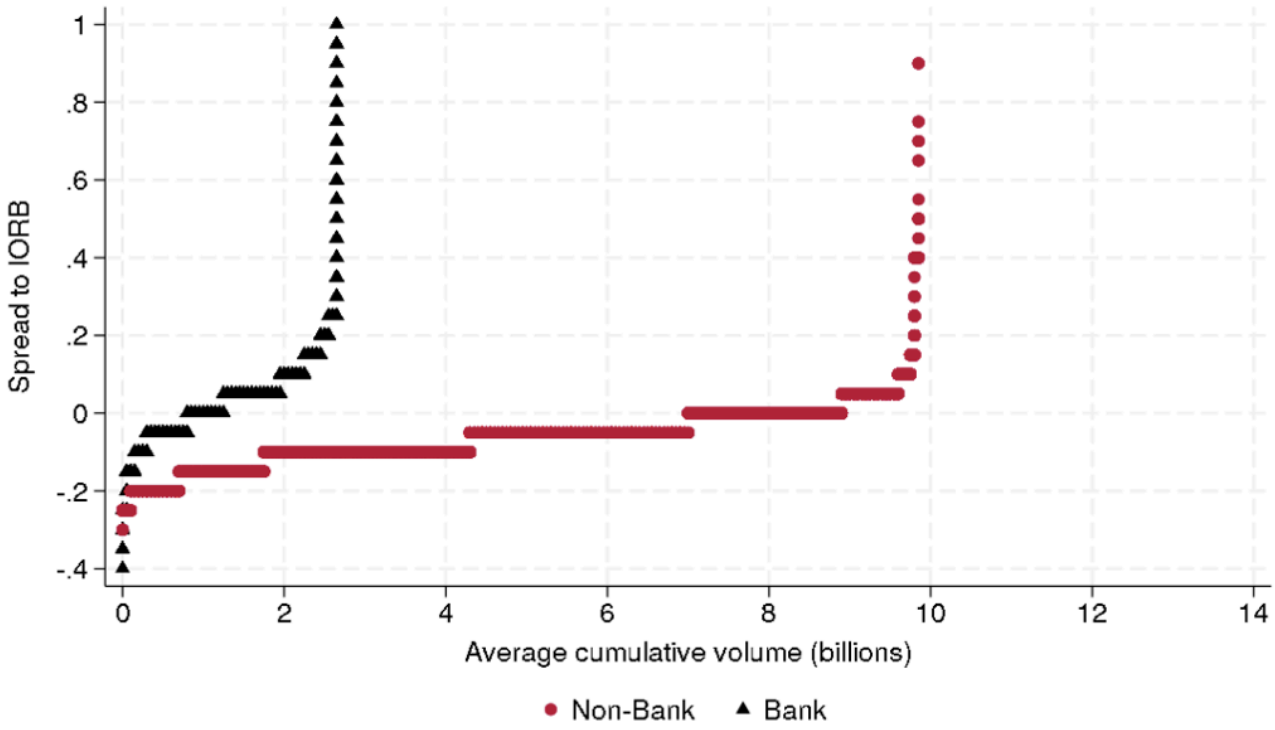
(i) Lender heterogeneity: banks vs. non-banks
Federal Home Loan Banks (FHLBs) are the dominant lenders, providing over 90% of fed funds lending. FHLBs cannot earn IORB at the Fed, incentivizing them to lend in the fed funds market. They supply funds elastically at rates below IORB to earn a positive return on otherwise idle balances. FHLBs' lending decisions depend on their regulatory liquidity buffers and alternative short-term investment opportunities.

(ii) Borrower heterogeneity: foreign vs. domestic banks
U.S. branches of foreign banks account for approximately 90% of total borrowing volume in the fed funds market. These foreign banks face fewer regulatory constraints, allowing them to

expand their balance sheet, borrowing from FHLBs at relatively lower rates and earning the spread to IORB more freely, an activity often termed the "Fed arb".

In contrast, domestic banks primarily borrow in this market to address genuine liquidity needs. They typically pay spreads around or slightly above IORB to obtain fed funds. Figure 1 shows the average cumulative borrowing of domestic banks in the fed funds market relative to the average spread they pay relative to IORB, separated by whether the lender is a bank or an FHLB. While there is some activity below minus 2 bps, the majority of trades are at spreads near zero or above, suggesting that domestic banks' participation is mainly driven by their need to source liquidity rather than arbitrage between FHLBs and IORB. Moreover, as we discuss below, smaller domestic banks are likely to engage intermittently in this market to manage transient liquidity shocks.

Figure 1. Rates and Volumes Borrowed by Domestic Banks in the Fed Funds Market



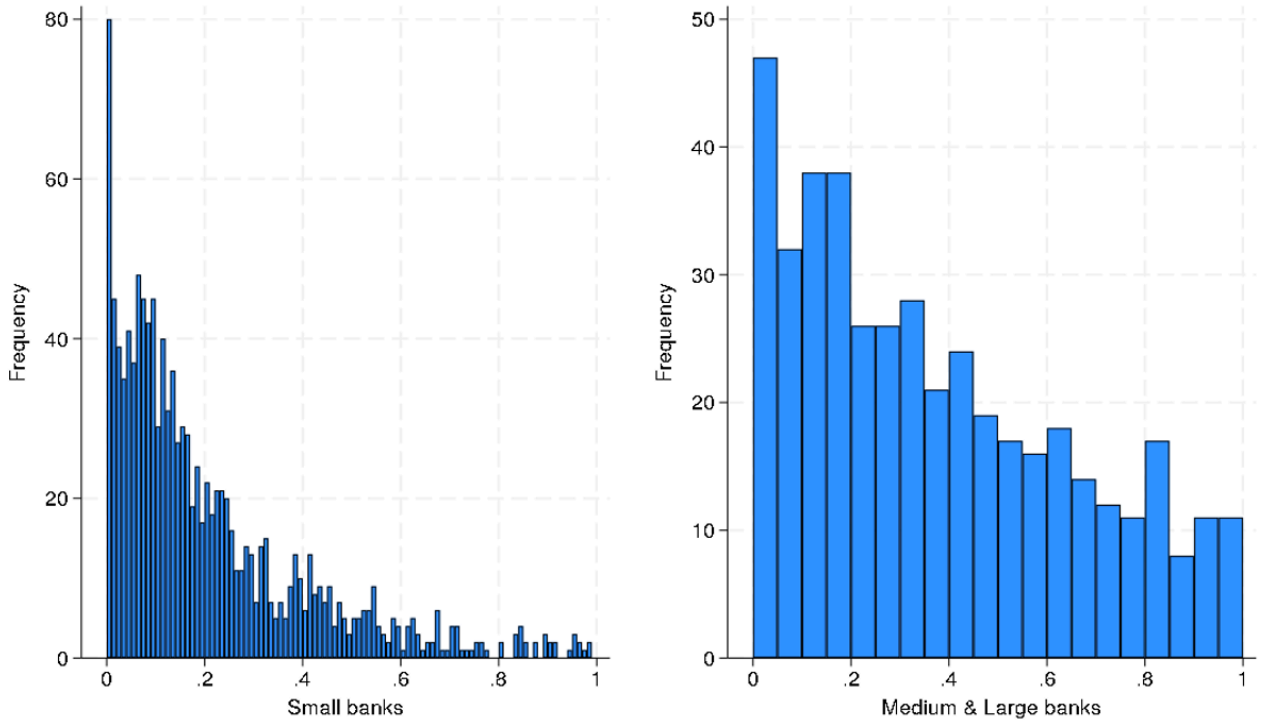
Note: This figure displays the relationship between cumulative average trading volumes and spreads to IORB in the fed funds market, separated by lender type. Each volume/spread point reflects at least 7 domestic banks' trading behavior averaged across our sample between October 2015 and January 2024. Black triangles represent cumulative trading volume lent by bank lenders, while red circles represent cumulative trading volume lent by non-bank lenders. This figure does not single out trading behavior on a particular day but represents average trading patterns over the entire sample period.
Source: FR 2420 Report of Selected Money Market Rates.
Accessible version

(iii) Size-based differences among domestic banks

Banks with assets over $100 billion maintain higher reserve-to-deposit ratios. These institutions are less sensitive to deposit fluctuations and rarely need to borrow in the fed funds market for

liquidity management. However, banks with assets under $10 billion show a tightly skewed distribution of reserve-to-deposit ratios. Figure 2 highlights the differences through histograms of the reserves-to-deposit rations of both cohorts. We see that small banks' distribution is heavily skewed to the left compared with the distribution of medium and large banks, suggesting that small banks are more likely to exhibit greater sensitivity to deposit outflows and are likely to adjust their reserves in response to deposit changes. Anbil, Infante, and Senyuz (2026) find that a $100 increase in weekly deposits at small institutions is associated with a $16 increase in reserves, compared to a much more modest response of larger domestic banks.

Figure 2. Reserves-to-Deposits Across Domestic Banks



Note: Histograms of domestic banks' reserves to deposit rations, separated by small banks (less than $10 billion in assets), and medium and large sized banks (greater than $10 billion).
Source: FR 2900 Report of Deposits and Vault Cash, internal Federal Reserve accounting records.
Accessible version

(iv) Banks' inelastic supply, especially as reserves decline

Anbil, Infante, and Senyuz (2026) reveals striking differences in supply elasticity between bank and non-bank lenders. In that paper, we find that non-bank lenders, primarily FHLBs, exhibit highly elastic supply behavior, while banks provide reserves inelastically.

To put the difference in elasticity into perspective, we find that on average, a $10 billion increase in weekly demand deposits at small domestic banks which have assets under $10 billion, is associated with a 141% increase in fed funds borrowing from non-bank lenders with rates remaining largely the same. In contrast, the same shock leads to a 77% increase in fed funds

borrowing from bank lenders, with a 16 basis point increase in rates charged by bank lenders, with little change in their lending volumes—a fairly inelastic sensitivity.

We further show that this inelasticity becomes more pronounced as aggregate reserves in the system decline. We estimate that bank lender elasticity almost doubles from 0.15 at high reserve levels to about 0.30 at the lowest observed reserve ratio, reflecting that banks become increasingly price-sensitive when liquidity is tight. By contrast, the non-bank elasticity remains essentially flat and statistically indistinguishable from zero, confirming that FHLBs and other non-bank suppliers continue to supply reserves at near-constant spreads regardless of aggregate reserve conditions. This stark divergence underscores that bank lenders drive the inelastic price response in the fed funds market as reserve levels come down.

(v) The role of bankers' banks

Bankers' banks serve as key intermediaries for smaller institutions, facilitating their access to wholesale funding markets.[2] They maintain larger reserve buffers relative to assets compared to typical commercial banks and provide their member banks access to reserve balances and interbank payments. The reserve positions of bankers' banks are particularly informative about liquidity conditions among smaller institutions. Changes in bankers' banks' reserve ratios are strongly negatively correlated with the spreads charged by bank lenders in the fed funds market.

IV. The Fed Funds Market Provides a Window into Distributional Frictions

This heterogeneity in the fed funds market that we described has important implications for interpreting market signals and understanding liquidity conditions. The behavior of different participant groups provides nuanced information about reserve distribution and potential stress points in the banking system.

While measures based on aggregate reserve levels are useful for broad assessments of system-wide liquidity, they often mask important distributional frictions. The interbank segment of the fed funds market offers valuable insights into true liquidity conditions that may not be apparent from headline figures. Bankers' banks play a crucial role in revealing these distributional frictions, as their activities reflect the liquidity needs of numerous smaller institutions. Recognizing these distinctions is crucial for policymakers and researchers seeking to assess the true state of liquidity in the financial system and the potential impacts of monetary policy decisions.

As the fed funds market dynamics have evolved since the GFC, interpreting aggregate measures like the effective federal funds rate (EFFR) has become more complicated, as this rate now reflects a mix of arbitrage activity and genuine liquidity management. In this environment, it is important to study granular data to accurately assess liquidity conditions across the banking system.

V. Conclusion

The federal funds market has undergone significant structural changes since the 2008 financial crisis, yet it retains its importance as a window into the functioning of the U.S. banking system. The key insight is that who lends matters as much as how much. Despite its reduced size, the

fed funds market remains a critical barometer for bank liquidity conditions. By providing insights into distributional frictions and liquidity conditions that are not captured by aggregate measures, the market continues to inform monetary policy decisions and financial stability assessments. These distributional frictions are bound to become more salient if a reduction in central bank reserves increases domestic banks' incentives to manage their liquidity in the fed funds market. The market's ability to reveal distributional frictions makes it an invaluable tool for policymakers and researchers alike. In an environment with less reserves in the banking system, these insights are likely to become even more important. As the financial landscape evolves, continued monitoring and analysis of the fed funds market will be essential for maintaining effective monetary policy implementation and ensuring the smooth functioning of the financial system.

References

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Anbil, S., Infante, S., & Senyuz, Z. (2026). A Tale of Demand and Supply for Central Bank Reserves. FEDS Working Paper.

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1. See Anderson & Na (2024) for more details on the evolution of the fed funds market since the GFC. Return to text
2. See Anbil, Anderson, Eyal, 2026 for a detailed description of bankers' banks and their role in the fed funds market. Return to text

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